

## CAMBRIDGE DORCHESTER, MD, USA

WMO: 722218

Lat: 38.539N

Lon: 76.030W

Elev: 6

StdP: 101.25

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 00356

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |              |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|--------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%          |            |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP           | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-7.9 | (c)<br>-6.3 | (d)<br>-17.6                  | (e)<br>0.8 | (f)<br>-6.1 | (g)<br>-15.0 | (h)<br>1.0 | (i)<br>-4.4 | (j)<br>11.0           | (k)<br>-0.2 | (l)<br>10.0 | (m)<br>-0.3 | (n)<br>3.3            | (o)<br>310 | (p)<br>0.420 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>9.4             | (c)<br>32.9     | (d)<br>24.8 | (e)<br>32.3 | (f)<br>24.4 | (g)<br>31.0 | (h)<br>23.8 | (i)<br>26.5         | (j)<br>31.0 | (k)<br>25.6 | (l)<br>29.8 | (m)<br>24.9 | (n)<br>28.9 | (o)<br>2.9           | (p)<br>210 |

| Dehumidification DP/MCDB and HR |             |             |             |             |             |             |             |             | Enthalpy/MCDB |             |             |             |             |             | Extreme<br>Max WB |
|---------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------|-------------|-------------|-------------|-------------|-------------|-------------------|
| 0.4%                            |             |             | 1%          |             |             | 2%          |             |             | 0.4%          |             | 1%          |             | 2%          |             |                   |
| DP                              | HR          | MCDB        | DP          | HR          | MCDB        | DP          | HR          | MCDB        | Enth          | MCDB        | Enth        | MCDB        | Enth        | MCDB        |                   |
| (a)<br>25.2                     | (b)<br>20.3 | (c)<br>28.4 | (d)<br>24.7 | (e)<br>19.8 | (f)<br>28.0 | (g)<br>23.6 | (h)<br>18.5 | (i)<br>27.2 | (j)<br>82.4   | (k)<br>30.6 | (l)<br>78.6 | (m)<br>30.0 | (n)<br>75.4 | (o)<br>29.5 | (p)<br>29.3       |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 8.8               | 7.8  | 7.0 | -12.3                      | 35.2 | 2.5                | 1.4 | -14.2                                              | 36.2 | -15.7      | 37.0 | -17.1      | 37.8 | -18.9      | 38.8 |
|                   |      |     | -12.1                      | 27.3 | 2.9                | 1.0 | -14.2                                              | 28.0 | -15.9      | 28.5 | -17.5      | 29.1 | -19.7      | 29.8 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 14.0   | 2.1   | 3.7   | 7.3   | 13.4  | 18.5  | 22.9  | 25.4  | 24.2  | 20.9  | 15.4  | 8.9   | 5.0   |  |  |
|                                                                        | DBStd       | 9.08   | 5.04  | 5.07  | 4.91  | 4.70  | 4.05  | 3.13  | 2.66  | 2.55  | 3.15  | 4.54  | 4.28  | 4.68  |  |  |
|                                                                        | HDD10.0     | 817    | 252   | 187   | 115   | 19    | 0     | 0     | 0     | 0     | 0     | 6     | 69    | 168   |  |  |
|                                                                        | HDD18.3     | 2291   | 505   | 410   | 342   | 160   | 50    | 3     | 0     | 0     | 11    | 114   | 282   | 413   |  |  |
|                                                                        | CDD10.0     | 2284   | 5     | 9     | 32    | 121   | 264   | 387   | 476   | 440   | 326   | 173   | 38    | 13    |  |  |
|                                                                        | CDD18.3     | 719    | 0     | 0     | 1     | 11    | 57    | 140   | 218   | 182   | 87    | 23    | 1     | 0     |  |  |
|                                                                        | CDH23.3     | 6253   | 0     | 0     | 4     | 102   | 406   | 1233  | 2156  | 1582  | 628   | 142   | 1     | 0     |  |  |
|                                                                        | CDH26.7     | 1946   | 0     | 0     | 0     | 19    | 77    | 379   | 812   | 487   | 147   | 26    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.7    | 3.2   | 3.2   | 3.5   | 3.3   | 2.6   | 2.2   | 1.9   | 1.8   | 2.2   | 2.6   | 2.8   | 2.8   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1183   | 92    | 81    | 105   | 89    | 111   | 91    | 124   | 120   | 97    | 83    | 90    | 92    |  |  |
|                                                                        | PrecMax     | 1698   | 192   | 198   | 204   | 180   | 249   | 297   | 314   | 273   | 264   | 203   | 217   | 226   |  |  |
|                                                                        | PrecMin     | 805    | 11    | 14    | 46    | 16    | 21    | 14    | 35    | 13    | 14    | 1     | 12    | 15    |  |  |
|                                                                        | PrecStd     | 221    | 38    | 45    | 43    | 38    | 61    | 61    | 64    | 78    | 63    | 49    | 53    | 56    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 17.5  | 20.1  | 23.7  | 28.8  | 31.1  | 34.9  | 36.2  | 33.0  | 32.8  | 29.0  | 22.2  | 19.9  |  |  |
|                                                                        |             | MCWB   | 14.7  | 15.5  | 17.0  | 20.1  | 23.0  | 24.7  | 26.8  | 25.0  | 23.8  | 22.6  | 16.6  | 17.0  |  |  |
|                                                                        | 2%          | DB     | 15.1  | 17.1  | 21.0  | 26.1  | 28.8  | 32.5  | 33.6  | 32.4  | 30.2  | 27.1  | 20.0  | 17.2  |  |  |
|                                                                        |             | MCWB   | 12.9  | 13.6  | 15.2  | 18.1  | 22.0  | 23.9  | 25.7  | 24.7  | 23.1  | 21.3  | 17.0  | 15.6  |  |  |
|                                                                        | 5%          | DB     | 12.6  | 14.1  | 17.7  | 23.7  | 27.2  | 31.0  | 32.4  | 31.1  | 27.9  | 25.0  | 17.4  | 15.1  |  |  |
|                                                                        |             | MCWB   | 10.3  | 10.9  | 13.4  | 17.6  | 20.5  | 23.2  | 24.9  | 24.1  | 21.6  | 19.9  | 14.5  | 13.0  |  |  |
|                                                                        | 10%         | DB     | 10.1  | 12.0  | 15.9  | 22.1  | 25.9  | 28.8  | 31.1  | 29.1  | 27.1  | 22.6  | 16.1  | 12.6  |  |  |
|                                                                        |             | MCWB   | 7.9   | 8.9   | 12.0  | 16.4  | 19.6  | 22.4  | 24.3  | 23.2  | 21.0  | 18.6  | 13.2  | 10.7  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 16.1  | 16.8  | 18.2  | 21.7  | 24.0  | 26.8  | 27.9  | 27.2  | 25.2  | 24.2  | 19.0  | 17.7  |  |  |
|                                                                        |             | MCDB   | 17.1  | 18.6  | 21.6  | 25.9  | 28.4  | 31.9  | 33.0  | 31.3  | 30.0  | 27.2  | 20.9  | 18.3  |  |  |
|                                                                        | 2%          | WB     | 13.5  | 14.4  | 15.9  | 19.6  | 22.9  | 25.2  | 26.9  | 26.0  | 24.3  | 22.6  | 17.2  | 15.7  |  |  |
|                                                                        |             | MCDB   | 15.4  | 16.6  | 19.0  | 24.4  | 27.1  | 29.5  | 31.8  | 29.7  | 28.5  | 25.4  | 18.6  | 16.8  |  |  |
|                                                                        | 5%          | WB     | 11.0  | 11.7  | 14.3  | 18.4  | 21.8  | 24.2  | 26.0  | 25.3  | 23.4  | 21.2  | 15.5  | 13.7  |  |  |
|                                                                        |             | MCDB   | 12.2  | 13.8  | 17.4  | 23.0  | 25.7  | 29.1  | 30.5  | 29.0  | 26.8  | 23.6  | 17.2  | 15.2  |  |  |
|                                                                        | 10%         | WB     | 7.8   | 9.1   | 12.6  | 16.9  | 21.0  | 23.4  | 25.2  | 24.5  | 22.4  | 19.7  | 13.9  | 11.2  |  |  |
|                                                                        |             | MCDB   | 9.4   | 11.6  | 15.0  | 20.7  | 24.3  | 27.7  | 29.4  | 28.2  | 25.5  | 22.9  | 16.1  | 12.2  |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 7.8   | 8.5   | 9.1   | 10.6  | 9.8   | 9.6   | 9.4   | 9.4   | 9.8   | 9.8   | 9.1   | 8.1   |  |  |
|                                                                        | 5% DB       | MCDBR  | 11.6  | 12.4  | 12.7  | 13.0  | 10.7  | 10.9  | 10.5  | 10.2  | 11.3  | 11.2  | 10.8  | 10.3  |  |  |
|                                                                        |             | MCWBR  | 10.0  | 9.4   | 8.3   | 6.9   | 5.3   | 4.5   | 4.4   | 4.3   | 5.3   | 6.3   | 7.8   | 9.2   |  |  |
|                                                                        | 5% WB       | MCDBR  | 10.8  | 11.7  | 11.7  | 11.7  | 9.4   | 9.7   | 9.6   | 8.8   | 9.6   | 9.1   | 8.7   | 9.5   |  |  |
|                                                                        |             | MCWBR  | 10.3  | 9.9   | 8.7   | 7.4   | 5.2   | 4.5   | 4.5   | 4.2   | 5.2   | 6.0   | 7.6   | 9.3   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.313 | 0.320 | 0.356 | 0.396 | 0.441 | 0.488 | 0.516 | 0.508 | 0.413 | 0.360 | 0.327 | 0.312 |  |  |
|                                                                        | taud        |        | 2.484 | 2.459 | 2.405 | 2.312 | 2.218 | 2.132 | 2.069 | 2.077 | 2.320 | 2.469 | 2.520 | 2.527 |  |  |
|                                                                        | Ebn at Noon |        | 866   | 908   | 904   | 882   | 845   | 802   | 775   | 770   | 832   | 850   | 844   | 842   |  |  |
|                                                                        | Edh at Noon |        | 79    | 93    | 108   | 126   | 141   | 154   | 163   | 158   | 117   | 91    | 77    | 72    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.17  | 3.01  | 4.01  | 5.09  | 5.78  | 6.34  | 6.17  | 5.51  | 4.48  | 3.38  | 2.45  | 1.91  |  |  |
|                                                                        | RadStd      |        | 0.17  | 0.32  | 0.34  | 0.36  | 0.65  | 0.48  | 0.29  | 0.33  | 0.46  | 0.36  | 0.21  | 0.18  |  |  |

## Historical Trends

|                              |  | DBAvg |     | Heating |        | Cooling |       | Degree-Days |         |         |         |
|------------------------------|--|-------|-----|---------|--------|---------|-------|-------------|---------|---------|---------|
|                              |  |       |     | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP       | HDD10.0 | HDD18.3 | CDD18.3 |
|                              |  | (d)   | (e) | (f)     | (g)    | (h)     | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A | N/A     | N/A    | N/A     | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (86 neighbors) |  | +0.45 | N/A | N/A     | N/A    | N/A     | N/A   | N/A         | N/A     | +139    | +84     |

Nomenclature: See separate page